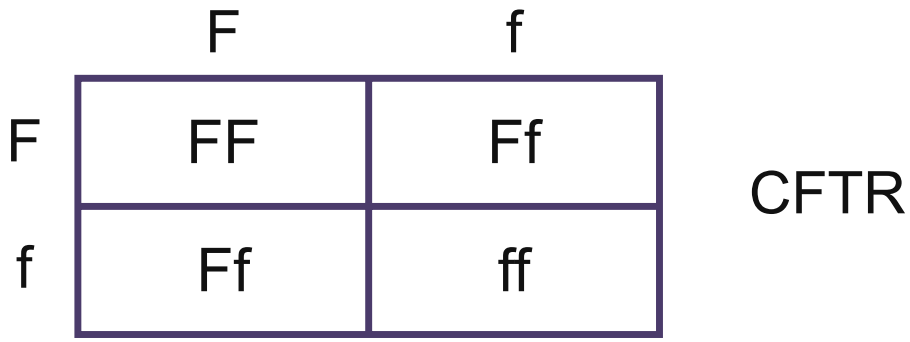
Reference A CFTR cross



Two carrier parents can each pass F or f. The possible pairs are FF, Ff, Ff and ff.

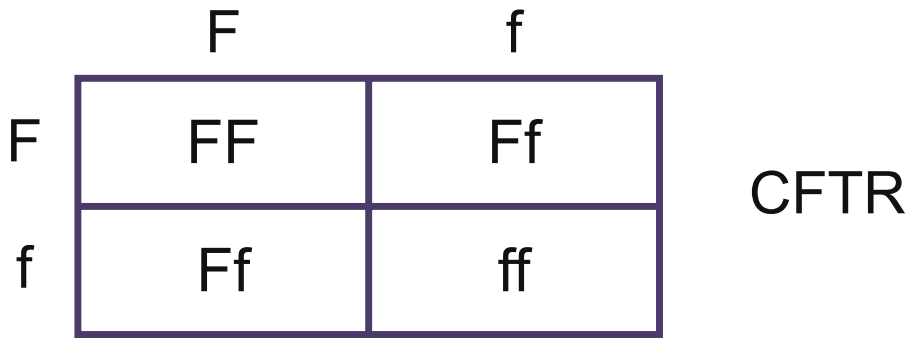
Reference only. F represents a usual CFTR copy and f a CF-causing variant in this simplified model.

Pair	Meaning in the model
FF	Two usual copies
Ff	One usual copy and one changed copy
ff	One changed copy from each parent

The model shows possibilities for each pregnancy in this cross. Use fictional examples.

Made by Matt

Reference B source card



The CFTR cross shows four possible pairs.

Reference only. Name the card and point to the line that supports the claim.

Section	Information
What it is	Cystic fibrosis is inherited and can affect breathing and digestion.
Inheritance	In the model, two CF-causing CFTR variants can be passed on, one from each parent. One changed copy usually means a carrier.
Care	Treatment can help manage symptoms. Infection precautions are a separate issue.
Source	NHS Cystic fibrosis card and the Genomics Education Programme inheritance notes.

Presentation marks

Criterion	Award one point when the presentation
Accurate fact	gives an accurate scientific fact
Named source	names the source so it can be identified
Inheritance	explains one gene copy from each parent

Made by Matt

Arrival and exit Aspire

Use the picture on Reference A for question 1 and the model table for questions 2–4.

1. Which two cells contain one F and one f?

2. Which gene do the letters represent?

3. How many changed copies does Ff show in the model?

4. Does the grid predict a named person’s outcome?

Exit ticket

1. Apply the mark scheme to a talk with an unsupported care claim. Justify the score.

2. Give a specific next step and demonstrate the repair.

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

Made by Matt

You do Aspire

Use your book or this sheet. Use Reference A and B. One response per task.

1. Present a short explanation that separates the inheritance model from a personal prediction.
Staff I ☐ S ☐ C ☐

---

2. Apply the mark scheme to a talk with an unsupported care claim. Justify the score.
Staff I ☐ S ☐ C ☐

---

3. Give a specific next step and demonstrate the repair.
Staff I ☐ S ☐ C ☐

Feedback and response · Aspire

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt